

No. of Experiment:

Name of the Experiment: “Determination of the specific resistance (resistivity) of a wire by metre bridge”.

Theory: The amount of resistance or hindrance that a conductor having unit length and unit area offers to the flow of current is called specific resistance of that conductor. It is denoted by ρ .

If the length of an wire is L , area of cross-section is A and total resistance is Q , then its specific resistance is

$$\rho = \frac{QA}{L}$$

If the wire is cylindrical or circular then its area of cross-section, $A = \pi r^2$

Here, r = radius of the wire.

Specific resistance,

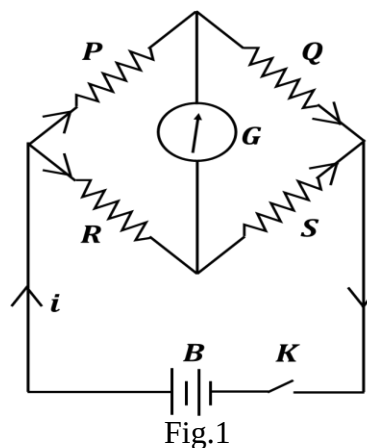
$$\rho = \frac{Q\pi r^2}{L} \quad \dots \dots \dots (1)$$

If R is expressed in practical unit and L and r are expressed in metre, then the unit of ρ will be ohm-metre ($\Omega - m$).

The total resistance Q in the equation (1) can be determined by a metre bridge according to the following theory.

In the left of the two gaps of the metre bridge a known resistance P and in the right gap an unknown resistance Q are inserted [Fig. 1] and after completing the circuit if the distance of the null point is l from the left terminal and if σ is the resistance of the unit length of the wire, then applying wheatstone principle we get,

$$\frac{P}{Q} = \frac{R}{S} = \frac{\text{Resistance of length } l \text{ of the bridge wire}}{\text{Resistance of length } (100 - l) \text{ of the bridge wire}}$$



$$\text{or, } \frac{P}{Q} = \frac{R}{S} = \frac{l\sigma}{(100-l)\sigma} = \frac{l}{(100-l)}$$

$$\text{or, } Q = \frac{P \times (100-l)}{l} \quad (\text{ohm}) \quad \dots \dots \dots (2)$$

Again, by changing the position of the resistances i.e., known resistance P in the right gap and unknown resistance Q in the left gap, we get

$$\frac{Q}{P} = \frac{l}{(100-l)}$$

$$\text{or, } Q = \frac{P \times l}{(100-l)} \quad (\text{ohm}) \quad \dots \dots \dots (3)$$

From the average value of the equations (1) and (2) we can find out the unknown resistance Q .

By measuring the radius r , length L of the wire and by determining the unknown resistance by the metre bridge and then inserting these values in equation (3) we can find out the value of the specific resistance of the wire.

Description of the wire:

It consist of three thick copper or brass strips XAM , NCH and LDY mounted on a long wooden board [Fig.2]. The resistances of these strips are negligible and there is connecting screw at each point of A, M, N, C, H, L and D .

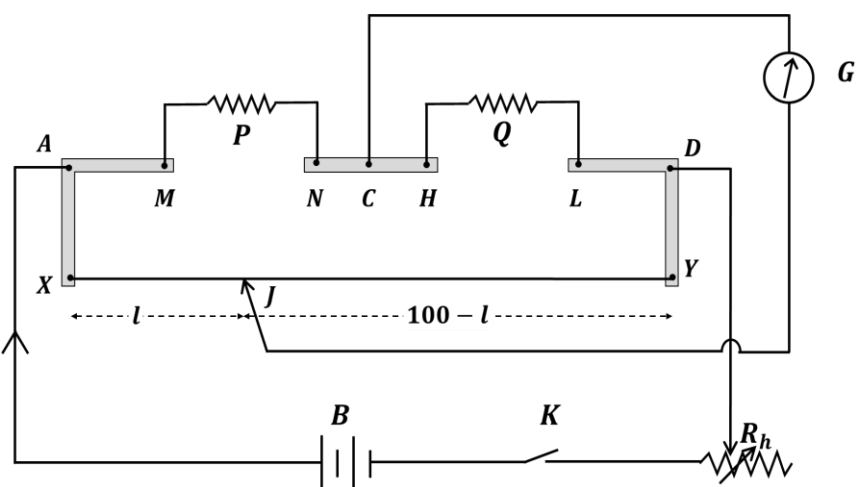


Fig.2

There is a metre scale placed between the strips XAM and LDY . A straight wire of $1m$ long and of uniform cross-section is attached at point X and Y along the length of the scale. $1m$ long wire is used as resistor, hence it is called a **metre bridge**.

List of Instruments:

1. A metre bridge
2. Electric cell
3. Galvanometer
4. Resistance box
5. Commutator or key
6. Jockey
7. Connecting wires
8. Screw gauge
9. Metre scale and
10. Experimental wire etc.

Procedure:

1. The circuit is drawn in a rough note book.
2. Connection is made according to the circuit diagram.
3. The unknown resistance connected in the right gap and the known resistance P is connected in the left gap of the bridge. The galvanometer is connected from the mid-point of the metre bridge and through the resistance box to the jockey J or a push key.
4. After completing the circuit connection the jockey is touched at one end of the wire of the bridge and the deflection of the galvanometer is observed. Again the jockey is touched at the other end of the metre bridge and deflection is noticed. If these two deflections are found opposite then it is considered that the circuit connection is alright.
5. A known resistance P_1 is applied in the circuit from the resistance box. Now holding the jockey on the wire it is moved right and left and a position on the wire is fixed whereat the galvanometer does not show any deflection. This is null point. The length l_1 is determined from the left of the meter scale attached to the bridge. Inserting these values in equation (2) the value of Q is found out.

6. Now positions of P and Q are interchanged. Keeping the value of P_1 fixed following the above procedure the null point is found out and the length l_1 upto that point is noted. These values are inserted in equation (3) and Q_1 is found out.
7. By interchanging the positions of P and Q are to the earlier position. A resistance P_2 of different value is withdrawn from the box P and the following procedures (5) and (6) unknown resistances Q_2 and Q'_2 are found out.
8. Following the procedure (7) resistance P_3 is taken out from the resistance box P and following the previous procedures Q_3 and Q'_3 are found out.
9. Now taking the average values of Q_1, Q'_1, Q_2, Q'_2 and Q_3, Q'_3 the unknown resistance Q ohm is found out.
10. By removing the experimental wire from the metre bridge its average length is measured by a metre scale.
11. Diameter i.e., radius of these wire is measured by a screw gauge and cross-sectional area of the wire is determined.
12. Inserting the experimental values of Q, L and r in equation (2), specific resistance of the wire is calculated.

Observation and Manipulation:

Pitch of the screw gauge = (mm)

Total no. of circular division of the screw gauge =

Least count of the screw gauge = $\frac{\text{Pitch}}{\text{Total no. of division of the circular scale}}$ = (mm)

Table 1: For the determination of resistance of the experimental wire:

No. of Obs.	Known resistance = P (ohm)	Position of the resistance, P	Distance of the null point = l (m)	Unknown resistance $Q = \frac{P \times (100 - l)}{l}$ (ohm)	Resistance, Q (ohm)	Average resistance, Q (ohm)
1.		Left gap				
2.						
3.						
1.		Right gap				
2.						
3.						

Table 2: For the determination of radius of the experimental wire:

No. of Obs.	Main scale reading = M (mm)	Circular scale reading = V	Least count = K (mm)	Fractional part, $F = V \times K$ (mm)	Total reading, $d' = M + F$ (mm)	Average reading, d (mm)	Radius, $r = d/2$ (mm)	Average radius, r (m)
1.								
2.								
3.								

Table 3: For the determination of specific resistance of experimental wire:

Length of the wire, L (m)	Total resistance of the wire, Q (ohm)	Cross-sectional area of the wire, $A = \pi r^2$ (m^2)	Specific resistance, $\rho = \frac{Q\pi r^2}{L}$ (Ωm)

Calculation:

$$\rho = \frac{Q\pi r^2}{L} = \dots \dots \dots (\Omega m)$$

Result:

Specific resistance of the experimental wire, $\rho = \dots \dots \dots (\Omega m)$

Precautions:

1. All connections are to be made tight.
2. Terminals of all connecting wires should be rubbed with sand paper.
3. Position of the null point is to be determined with special care.
4. For getting correct value of unknown resistance end correction of the metre bridge is to be determined.
5. Radius of wire is to be determined correctly.
6. Value of ρ is to be determined correctly.